

Notes: Fishman and Hagerty (JF, 1989)
Disclosure Decisions by Firms and the Competition for Price Efficiency

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Model environment and key objects	4
3.1	Timing	4
3.2	Firms and investment	4
3.3	Disclosure technology	4
3.4	Traders and market maker	5
4	Securities market equilibrium and price efficiency	5
4.1	Linear trading and pricing equilibrium	5
4.2	Price as a weighted average	5
4.3	Interpretation of price efficiency	5
5	Investment incentives	6
5.1	Manager's investment problem	6
5.2	Why better prices increase investment	6
5.3	Firm value from disclosure and trader attention	6
6	Disclosure decisions and the competition for trader attention	7
6.1	Trader payoffs	7
6.2	Equilibrium allocation of traders	7
6.3	Firm disclosure equilibrium	7
7	Welfare and interpretation	8
7.1	Social surplus	8
7.2	Overdisclosure proposition	8
7.3	Why firms may overdisclose	8
7.4	Asymmetric equilibria	8
8	Tables and figures	9

8.1	Figure 1: Disclosure equilibrium as firm risk varies	9
8.2	Figures 2 and 3: Disclosure equilibrium as trader supply varies	9
9	Short summary	10
10	Conclusion	11
10.1	Core conclusion	11
10.2	Economic intuition	11
10.3	Contribution revisited	11
10.4	Policy implication	11
10.5	Limitations	12
10.6	Takeaway	12

1 Big picture

- **Research question.** Why would firms voluntarily spend resources on disclosure, and can voluntary disclosure ever be excessive from a social point of view?
- **Core idea.** Disclosure makes security prices more informative. More informative prices improve managers' investment incentives because the market price becomes more sensitive to the manager's unobservable investment decision.
- **Key friction.** Public disclosure is not costless to use. Even if a disclosure is publicly available, traders must spend scarce attention and effort to understand it. Each trader can study only one firm's disclosure.
- **Competition for price efficiency.** Firms compete for traders' attention. A more informative disclosure attracts more informed traders, which makes the firm's own stock price more efficient.
- **Main result.** Firms can disclose more information than is socially optimal. The private benefit of disclosure includes stealing traders from other firms, while the social planner treats this as a reallocation rather than a net gain.
- **Contrast with standard disclosure policy concern.** Much mandatory-disclosure policy is motivated by the worry that firms will disclose too little. Fishman and Hagerty show a theoretical force in the opposite direction: firms may overdisclose.
- **Price efficiency and real efficiency.** The paper links financial-market efficiency to real investment efficiency. Prices are not just signals to outside investors; by reacting to managerial actions, they discipline real decisions.
- **Economic takeaway.** Disclosure can create a private race for investor attention. This race improves each firm's own price efficiency but can waste resources and reduce price efficiency elsewhere.

2 Incremental contribution of the paper

- **Disclosure as a way to improve investment incentives.** Earlier disclosure models often focus on how disclosure affects investors' information or reduces duplication in information production. This paper emphasizes that better prices can induce better real investment choices.
- **Assimilation costs, not just production costs.** The paper separates the cost of producing information from the cost traders incur to understand and use that information. The second cost is central to the overdisclosure result.
- **Attention competition across firms.** A firm's disclosure choice affects how traders allocate attention across all firms. This cross-firm allocation of attention generates an externality.
- **A voluntary overdisclosure result.** In contrast to the usual argument for mandatory disclosure based on underproduction of information, the model shows that private incentives can lead firms to spend too much on disclosure.
- **A real effect of market prices.** The paper provides a channel through which price efficiency affects real firm behavior even though managers do not directly learn from the price. What matters is that price becomes more sensitive to investment.

- **Foundation for later attention and governance models.** The model anticipates later work in which investor attention, information processing capacity, and market feedback affect corporate choices.

3 Model environment and key objects

3.1 Timing

The model has three dates.

- **Date 1.** Each manager chooses investment and disclosure precision. The manager also distributes remaining cash to current shareholders.
- **Date 2.** Traders choose which firm's disclosure to study, informed and liquidity traders submit orders, and a competitive market maker sets the stock price from total order flow.
- **Date 3.** The firm's terminal payoff is realized.

The manager maximizes current shareholders' expected wealth. Current shareholders liquidate their positions at Date 2, so the manager cares about the dividend plus the expected Date 2 stock price.

3.2 Firms and investment

There are I all-equity firms. Firm i has privately observed cash m_i and a project that pays θ_i at Date 3. The manager chooses the expected payoff level by investing $C(\bar{\theta}_i)$, where

$$C' > 0, \quad C'' > 0.$$

The investment decision is observed only by the manager. This unobservability creates underinvestment: if prices do not respond strongly to the manager's action, the manager does not fully internalize the benefit of investment in the price at which current shareholders sell.

3.3 Disclosure technology

The manager releases a signal

$$s_i = \theta_i + \varepsilon_i,$$

where the noise term has precision τ_i . Higher precision means a more informative disclosure. For tractability, disclosure precision is binary:

$$\tau_i \in \{\tau_L, \tau_H\}, \quad \tau_H > \tau_L > 0.$$

The cost of disclosure is $D(\tau_i)$, with

$$D(\tau_H) > D(\tau_L) \geq 0.$$

Interpretation. A high-precision disclosure can represent more detailed financial reporting, more extensive explanation of operations, more credible auditing, or better dissemination of information.

3.4 Traders and market maker

There are N risk-neutral informed traders. A trader can study one firm's disclosure at no direct cost, but studying additional firms is prohibitively costly. This creates a limited-attention environment.

If n_i traders study firm i , each observes the same signal s_i and submits a market order. Liquidity traders submit noise order flow z_i . A competitive, risk-neutral market maker observes total order flow and sets price equal to expected value conditional on that order flow.

Interpretation. The disclosure is public in the legal sense, but it is costly to process. The relevant scarce input is not access to the document; it is attention and analytical capacity.

4 Securities market equilibrium and price efficiency

4.1 Linear trading and pricing equilibrium

Given disclosure precision τ_i , number of informed traders n_i , and the market's conjecture about investment, the securities market has a unique symmetric linear equilibrium of the Kyle/Admati–Pfleiderer type:

$$\begin{aligned} X_i(s_i) &= \beta_{0i} + \beta_{1i}s_i, \\ P_i(n_ix_i + z_i) &= \lambda_{0i} + \lambda_{1i}(n_ix_i + z_i). \end{aligned}$$

Informed traders buy when the disclosure signal is high relative to the market's prior and sell when it is low. The market maker extracts information from order flow, but order flow is noisy because it mixes informed trading and liquidity trading.

4.2 Price as a weighted average

The equilibrium price can be written as a weighted average of the market's prior belief and the signal inferred from trading:

$$p_i = \bar{\theta}_i^c \frac{h_\theta}{h_\theta + \phi(\tau_i, n_i, h_\theta)} + \left(s_i + \frac{z_i}{n_i\beta_{1i}} \right) \frac{\phi(\tau_i, n_i, h_\theta)}{h_\theta + \phi(\tau_i, n_i, h_\theta)},$$

where

$$\phi(\tau_i, n_i, h_\theta) = \frac{\tau_i n_i h_\theta}{\tau_i + h_\theta + n_i h_\theta}.$$

The term $\phi(\tau_i, n_i, h_\theta)$ summarizes the precision of the trading signal about firm value. It increases when the disclosure is more precise and when more informed traders study the firm.

4.3 Interpretation of price efficiency

Price efficiency in this model means that the Date 2 stock price is more sensitive to the information about the firm's Date 3 payoff. Two forces increase this sensitivity:

- **Higher disclosure precision.** A more precise signal s_i gives informed traders a better informational advantage.

- **More informed traders.** More traders studying firm i make aggregate order flow more responsive to the signal and less dominated by liquidity trading noise.

The important point is that the price becomes more informative even though the manager does not learn from it. The price disciplines the manager because it affects the price received by current shareholders.

5 Investment incentives

5.1 Manager's investment problem

Taking τ_i and n_i as given, the manager chooses investment to maximize expected price minus investment and disclosure costs. The first-order condition is

$$\frac{\phi(\tau_i, n_i, h_\theta)}{h_\theta + \phi(\tau_i, n_i, h_\theta)} = C'(\bar{\theta}_i).$$

The left-hand side is the weight placed on the signal generated by trading. Since the signal is noisy, this weight is less than one. Therefore, the manager chooses investment below the first-best level, which would satisfy $C'(\bar{\theta}_i) = 1$.

5.2 Why better prices increase investment

The optimal investment level, denoted $\bar{\theta}^*(\tau_i, n_i)$, is increasing in $\phi(\tau_i, n_i, h_\theta)$. Therefore:

- higher disclosure precision increases investment;
- more informed traders studying the firm increase investment;
- greater price efficiency moves the investment decision closer to first best.

Economic intuition. If the market price barely responds to the manager's action, the manager has little reason to invest for current shareholders. If the price reacts strongly to the action through informed trading, investment is more strongly capitalized into the sale price.

5.3 Firm value from disclosure and trader attention

Substituting the optimal investment policy into expected shareholder value gives

$$V(\tau_i, n_i) = \bar{\theta}^*(\tau_i, n_i) - C(\bar{\theta}^*(\tau_i, n_i)) - D(\tau_i).$$

A firm benefits from greater disclosure precision if the investment-incentive gain exceeds the extra disclosure cost. It also benefits from having more informed traders study its disclosure, because that improves price efficiency without directly raising the firm's disclosure cost.

6 Disclosure decisions and the competition for trader attention

6.1 Trader payoffs

Before trading, informed traders choose which firm to study. Lemma 2 shows that a trader's expected profit from studying firm i is increasing in disclosure precision and decreasing in the number of other traders studying the same firm:

$$\pi(\tau_i, n_i) = \frac{1}{n_i + 1} \left\{ \frac{\tau_i}{n_i(h_\theta + \tau_i)h_\theta h_z} \right\}^{1/2}.$$

Interpretation. More precise disclosure makes informed trading more profitable. More traders studying the same firm make trading less profitable because the price becomes more responsive to the same information and competition among informed traders dissipates rents.

6.2 Equilibrium allocation of traders

In equilibrium, traders spread across firms until expected trading profits are equalized across the disclosures that are being studied. If all firms choose the same precision, traders distribute evenly:

$$n^*(\tau_L, 0) = n^*(\tau_H, I) = \frac{N}{I}.$$

If one firm chooses high precision while others choose low precision, more traders study that high-precision firm. More generally:

- a firm attracts more traders by choosing τ_H instead of τ_L ;
- a firm attracts more traders when fewer other firms choose τ_H ;
- trader attention is therefore a relative, competitive object.

This is the central strategic interaction in the paper. A firm can improve its own price efficiency partly by pulling traders away from other firms.

6.3 Firm disclosure equilibrium

Let I' denote the number of firms choosing high disclosure precision τ_H . A pure-strategy equilibrium with I' high-disclosure firms requires that neither high-disclosure firms nor low-disclosure firms want to switch. For $0 < I' < I$, the conditions are

$$V(\tau_L, n^*(\tau_L, I')) \geq V(\tau_H, n^*(\tau_H, I' + 1)),$$

and

$$V(\tau_H, n^*(\tau_H, I')) \geq V(\tau_L, n^*(\tau_L, I' - 1)).$$

The paper proves that there is always at least one pure-strategy equilibrium.

7 Welfare and interpretation

7.1 Social surplus

If I' firms choose high precision and $I - I'$ choose low precision, social surplus is

$$I'V(\tau_H, n^*(\tau_H, I')) + (I - I')V(\tau_L, n^*(\tau_L, I')).$$

Trading profits are not included because they are transfers among traders and other market participants. The social planner cares about real investment gains and disclosure costs, not about informed traders' redistributive profits.

7.2 Overdisclosure proposition

Let I^* be the number of high-disclosure firms that maximizes social surplus. Proposition 2 states that in any equilibrium,

$$I' \geq I^*.$$

Thus, firms' disclosure expenditures either equal or exceed the socially optimal amount.

7.3 Why firms may overdisclose

The overdisclosure result comes from a negative externality. When firm i chooses a more precise disclosure, it receives two private benefits:

- the direct benefit of a more informative signal;
- the indirect benefit of attracting more traders, which improves its own price efficiency.

But the second benefit is partly a business-stealing effect. Traders attracted to firm i are unavailable to study other firms. Other firms' prices become less efficient, and their investment incentives weaken. Individual firms ignore that cost.

Economic message. The private marginal benefit of disclosure can exceed the social marginal benefit because firms compete for a limited pool of informed attention.

7.4 Asymmetric equilibria

The paper also shows that in an asymmetric equilibrium, firms that spend more on disclosure have higher market values. This resembles standard voluntary-disclosure results in which better firms disclose more, but the causality is different here.

- In standard models, more valuable firms choose to disclose because they want to separate from less valuable firms.
- In this model, firms are ex ante identical; choosing more disclosure makes the firm more valuable by attracting traders and improving investment incentives.

Interpretation. Disclosure can create value by changing the market-information environment around the firm, not merely by revealing preexisting value.

8 Tables and figures

8.1 Figure 1: Disclosure equilibrium as firm risk varies

Read:

- Figure 1 studies a two-firm example with $\tau_L = 1$, $\tau_H = 2$, and $N/2 = 25$ traders observing each disclosure in a symmetric allocation.
- The horizontal axis is firm risk, measured as $1/h_\theta$.
- The figure plots firm value under low and high disclosure precision, $V(\tau_L, N/2)$ and $V(\tau_H, N/2)$.
- For low firm risk, the equilibrium is low disclosure by both firms, (τ_L, τ_L) .
- For intermediate risk, both firms choose high disclosure, (τ_H, τ_H) .
- For sufficiently high risk, the equilibrium returns to low disclosure, (τ_L, τ_L) .

Economic message: The disclosure choice is not monotone in firm risk. Higher price efficiency is more valuable when investment incentives are more important, but high disclosure also costs more and changes trader allocation.

8.2 Figures 2 and 3: Disclosure equilibrium as trader supply varies

Read Figure 2:

- Figure 2 varies the total number of traders N while holding firm risk fixed.
- When there are few traders, both firms choose low precision.
- For an intermediate range of traders, the equilibrium is asymmetric: one firm chooses high precision and the other low precision.
- When there are many traders, both firms choose high precision.

Read Figure 3:

- Figure 3 shows that the comparative statics can reverse under different parameter values.
- When there are few traders, both firms choose high precision.
- For an intermediate range of traders, there can be multiple equilibria.
- When there are many traders, both firms choose low precision.

Economic message: Trader attention is strategic. Adding more traders can change both the private value of disclosure and the intensity of firms' competition for attention. Equilibrium disclosure can therefore be symmetric, asymmetric, or multiple.

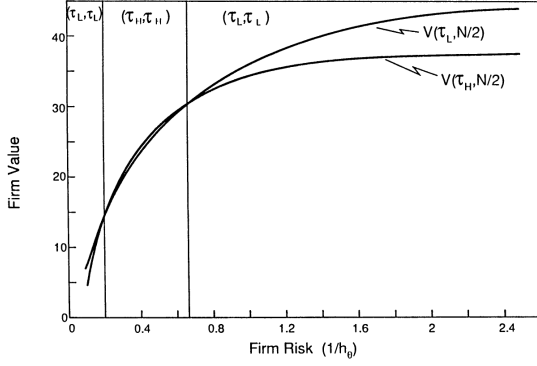


Figure 1: Two-firm example #1: firm value and disclosure equilibria as firm risk varies.

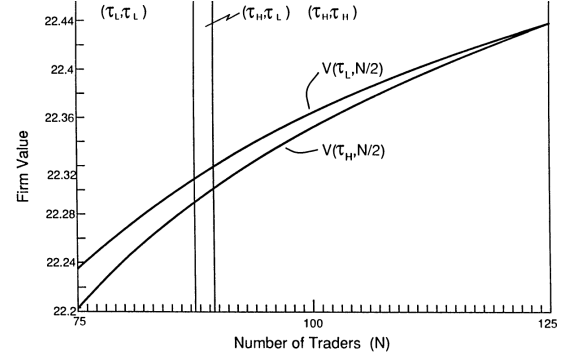


Figure 2: Two-firm example #2: firm value and disclosure equilibria as the number of traders varies.

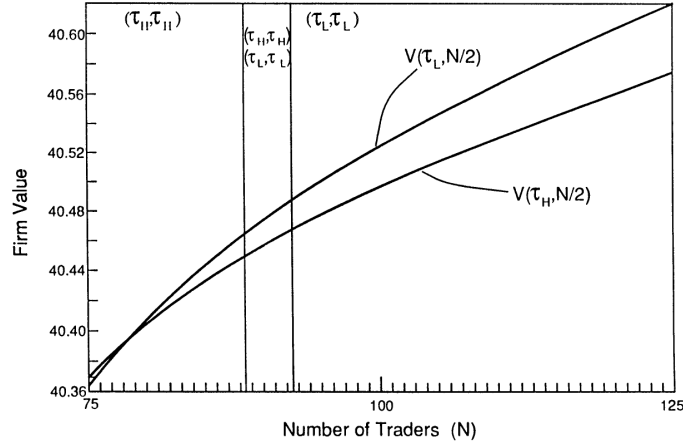


Figure 3: Two-firm example #3: a different parameterization in which the trader-comparative statics are nearly reversed.

9 Short summary

- The paper studies how firms choose disclosure when disclosure affects stock-price efficiency and price efficiency affects investment incentives.
- Managers choose investment that is not directly observed by the market.
- More informative stock prices make investment more valuable to current shareholders because the Date 2 sale price responds more strongly to investment.
- Firms can increase price efficiency by issuing more informative disclosures.
- Traders must decide which firms' disclosures to study, and each trader can study only one firm.
- A more informative disclosure attracts more traders, raising the firm's own price efficiency.

- This creates a negative externality: the firm attracts traders away from other firms and lowers their price efficiency.
- Because firms do not internalize this externality, equilibrium disclosure can exceed the socially optimal level.
- The result challenges the view that private disclosure incentives always generate too little disclosure.
- The model also predicts that, in asymmetric disclosure equilibria, firms with more disclosure have higher values even though firms are ex ante identical.

10 Conclusion

10.1 Core conclusion

Fishman and Hagerty show that voluntary disclosure can be privately excessive. Disclosure improves a firm's own stock-price efficiency and thereby improves investment incentives. But because traders have limited attention, one firm's disclosure can draw traders away from other firms. The firm internalizes the private price-efficiency gain but not the price-efficiency loss imposed elsewhere.

10.2 Economic intuition

The model turns disclosure into a competition for scarce informed attention. A firm wants traders to study its disclosure because informed trading makes its price more sensitive to its investment. But in equilibrium, attention spent on one firm cannot be spent on another. The social planner therefore sees part of the benefit of disclosure as merely a reallocation of attention, while individual firms treat it as a private gain.

10.3 Contribution revisited

The paper's main conceptual contribution is to connect three ideas: corporate disclosure, information-processing costs, and real investment incentives. It shows that price efficiency is valuable because it improves real decisions, but also that the decentralized pursuit of price efficiency can be inefficient.

10.4 Policy implication

The paper cautions against assuming that mandatory disclosure always fixes an underdisclosure problem. If firms already have incentives to disclose too much in order to compete for trader attention, additional mandatory disclosure can aggravate rather than solve the inefficiency. The result does not say all mandatory disclosure is harmful; it says the welfare effect depends on the costs of producing and processing information and on the cross-firm allocation of attention.

10.5 Limitations

The model is intentionally stylized. Disclosure precision is binary, traders can study only one firm, firms are ex ante identical in the numerical examples, and the paper abstracts from many real-world channels such as legal liability, proprietary costs, heterogeneous investors, and positive information spillovers across related firms. These simplifications make the mechanism transparent but limit direct empirical interpretation.

10.6 Takeaway

The enduring takeaway is that disclosure policy must account for both information production and information assimilation. More information can improve prices and investment, but when attention is scarce, firms may compete too intensely for price efficiency and disclose more than is socially optimal.